

Compare permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Note: Perform the exercise as a normal user. Do not use sudo.

Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400 (r--)	Try to read and modify the file	Read succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds if directory has r but not x? Correction: listing succeeds, entering fails because x is required.
200 (-w-)	Try to read and modify the file	Read fails; writing/modification succeeds (subject to how the program opens the file).	Try to list, enter, and create a file	Listing fails; entering fails; creating files fails because x is required to access directory entries.
100 (--x)	Try to read and execute the file	Reading fails; execution may succeed only if the file is a valid executable/script and the required interpreter/access conditions are met.	Try to list and enter the directory	Listing fails; entering succeeds.
500 (r-x)	Try to read and modify the file	Reading succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds.
600 (rw-)	Try to read and modify the file	Reading and modification succeed.	Try to list, enter, and create a file	Listing succeeds; entering fails; creating a file fails.
700 (rwx)	Try to read, modify, and execute the file	Reading and modification succeed; execution is allowed by the permission bits (assuming the contents are executable).	Try to list, enter, create, and delete files	All operations succeed.

```
Session Actions Edit View Help
(gopika@kali)-[~/linux/lab/file_permission]
$ ls
sample_dir  sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ ls -l
total 4
drwxrwxr-x 2 gopika gopika 4096 Aug 22 19:21 sample_dir
-rw-rw-r-- 1 gopika gopika   0 Aug 22 19:19 sample.txt
```

1. 400 (r--)

```
(gopika@kali)~/linux/lab/file_permission
$ ls
sample_dir  sample.txt

(gopika@kali)~/linux/lab/file_permission
$ ls -l
total 4
drwxrwxr-x 2 gopika gopika 4096 Aug 22 19:21 sample_dir
-rw-rw-r-- 1 gopika gopika   0 Aug 22 19:19 sample.txt

(gopika@kali)~/linux/lab/file_permission
$ chmod 400 sample.txt

(gopika@kali)~/linux/lab/file_permission
$ ls -l sample.txt
-r----- 1 gopika gopika   0 Aug 22 19:19 sample.txt

(gopika@kali)~/linux/lab/file_permission
$ cat sample.txt

(gopika@kali)~/linux/lab/file_permission
$ echo "Hello" > sample.txt
zsh: permission denied: sample.txt

(gopika@kali)~/linux/lab/file_permission
$
```

400 write permission is denied.

2. 200 (-w-)

```
(gopika@kali)~/linux/lab/file_permission
$ chmod 200 sample.txt

(gopika@kali)~/linux/lab/file_permission
$ chmod 200 sample_dir

(gopika@kali)~/linux/lab/file_permission
$ echo "hello gopika" >> sample.txt

(gopika@kali)~/linux/lab/file_permission
$ cat sample.txt
cat: sample.txt: Permission denied

(gopika@kali)~/linux/lab/file_permission
$ ls -ld sample.txt sample_dir
d-w----- 2 gopika gopika 4096 Aug 22 19:21 sample_dir
--w----- 1 gopika gopika   13 Aug 22 19:46 sample.txt

(gopika@kali)~/linux/lab/file_permission
$ cat sample.txt
cat: sample.txt: Permission denied

(gopika@kali)~/linux/lab/file_permission
$ echo "hello" > sample.txt

(gopika@kali)~/linux/lab/file_permission
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(gopika@kali)~/linux/lab/file_permission
$ cd sample_dir
cd: permission denied: sample_dir

(gopika@kali)~/linux/lab/file_permission
$ touch sample_dir/file2.txt
touch: cannot touch 'sample_dir/file2.txt': Permission denied
```

3. 100 (--x)

```
(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 100 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 100 sample_dir

(gopika@kali)-[~/linux/lab/file_permission]
$

(gopika@kali)-[~/linux/lab/file_permission]
$ ls -ld sample.txt sample_dir
d--x----- 2 gopika gopika 4096 Aug 22 19:21 sample_dir
--x----- 1 gopika gopika    6 Aug 22 19:53 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
cat: sample.txt: Permission denied

(gopika@kali)-[~/linux/lab/file_permission]
$ ./sample.txt
zsh: permission denied: ./sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(gopika@kali)-[~/linux/lab/file_permission]
$ cd sample_dir

(gopika@kali)-[~/linux/lab/file_permission/sample_dir]
$ pwd
/home/gopika/linux/lab/file_permission/sample_dir
```

4. 500 (r-x)

```
(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 500 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 500 sample_dir
chmod: cannot access 'sample_dir': No such file or directory

(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 500 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 500 sample_dir

(gopika@kali)-[~/linux/lab/file_permission]
$ ls -ld sample.txt sample_dir
dr-x----- 2 gopika gopika 4096 Aug 22 19:21 sample_dir
-r-x----- 1 gopika gopika    6 Aug 22 19:53 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
hello

(gopika@kali)-[~/linux/lab/file_permission]
$ echo "hii gopikaahoe ru " > sample.txt
zsh: permission denied: sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ ls sample_dir
file1.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cd sample_dir

(gopika@kali)-[~/linux/lab/file_permission/sample_dir]
$ pwd
/home/gopika/linux/lab/file_permission/sample_dir
```

5. 600 (rw-)

```
(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 600 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 600 sample_dir

(gopika@kali)-[~/linux/lab/file_permission]
$ ls -ld sample.txt sample_dir
drw----- 2 gopika gopika 4096 Aug 22 19:21 sample_dir
-rw----- 1 gopika gopika  16 Aug 22 20:10 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
hello gopika gm

(gopika@kali)-[~/linux/lab/file_permission]
$ echo "gm gopika" > sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
gm gopika

(gopika@kali)-[~/linux/lab/file_permission]
$ ls sample_dir
ls: cannot access 'sample_dir/file1.txt': Permission denied
file1.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cd sample_dir
cd: permission denied: sample_dir

(gopika@kali)-[~/linux/lab/file_permission]
$ touch sample_dir/file2.txt
touch: cannot touch 'sample_dir/file2.txt': Permission denied

(gopika@kali)-[~/linux/lab/file_permission]
$ touch sample_dir/file.txt
touch: cannot touch 'sample_dir/file.txt': Permission denied
```

6. 700 (rwx)



A terminal window showing a series of commands and their outputs. The user is in a directory `~/linux/lab/file_permission`. The commands and outputs are as follows:

```
(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 700 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ chmod 700 sample_dir

(gopika@kali)-[~/linux/lab/file_permission]
$ ls -ld sample.txt sample_dir
drwx----- 2 gopika gopika 4096 Aug 22 19:21 sample_dir
-rwx----- 1 gopika gopika  10 Aug 22 21:15 sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
gm gopika

(gopika@kali)-[~/linux/lab/file_permission]
$ echo "good nighth gopika" > sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
good nighth gopika

(gopika@kali)-[~/linux/lab/file_permission]
$ ./sample.txt: 1: goodnight gopika: not found
zsh: no such file or directory: ./sample.txt:

(gopika@kali)-[~/linux/lab/file_permission]
$ ./sample.txt
./sample.txt: 1: good: not found

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
good nighth gopika

(gopika@kali)-[~/linux/lab/file_permission]
$ echo "bad nighth gopika" > sample.txt

(gopika@kali)-[~/linux/lab/file_permission]
$ cat sample.txt
bad nighth gopika

(gopika@kali)-[~/linux/lab/file_permission]
$ ./sample.txt
./sample.txt: 1: bad: not found

(gopika@kali)-[~/linux/lab/file_permission]
$ cd sample_dir

(gopika@kali)-[~/linux/lab/file_permission/sample_dir]
$ touch new.txt
```

Questions:

1. What does read (r) permission allow you to do with a regular file?

It allows you to view or read the contents of the file.

2. What does read (r) permission allow you to do with a directory?

It allows you to view the contents of the directory, such as the list of filenames.

It does not by itself allow you to enter or access the files inside it.

3. What does write (w) permission mean for a regular file?

It allows you to modify, overwrite, or write data to the file.

It does not automatically allow you to read its contents.

4. What does write (w) permission mean for a directory?

It allows you to create, delete, or rename entries in the directory.

The execute (x) permission is normally also required to perform these operations.

5. What does execute (x) permission mean for a regular file?

It allows the file to be executed as a program or script, provided it is in a suitable executable format.

6. Why is execute (x) permission important for accessing a directory?

The x permission means search or traverse permission.

It allows you to enter the directory and access files or subdirectories inside it, when the required permissions are available.

7. Which permission combination allows the owner to fully access a regular file?

700 (rwx) gives the owner read, write, and execute permissions.

8. Which permission combination allows the owner to fully access a directory?

700 (rwx) gives the owner read, write, and execute (traverse) permissions.

This allows the owner to list, create, delete, and rename entries in the directory.

9. Why do the same r, w, and x permissions behave differently for a regular file and a directory?

A regular file and a directory are different types of filesystem objects, so the permissions have different meanings.

For a regular file:

r → Read the file contents

w → Modify the file contents

x → Execute the file

For a directory:

r → View the list of entries

w → Create, delete, or rename entries

x → Enter/traverse the directory and access its entries

Therefore, the same r, w, and x permissions behave differently because files store data, while directories organize and manage filesystem entries.